Convergence analysis of teacher-model OBS methods in LR

Matthew Ward and Kevin Miller

Abstract

1. Introduction

- Kaczmarz method
- Brief introduction to online sample (batch?) selection
- Justify reasoning behind teacher-informed methods
  - Real situations in which we might have a teacher available
  - Derivation of optimal criterion given perfect teacher
  - Define teacher-optimal criterion
- Summarize results of paper, especially computational results

1.1. Setup

With some $n, d \in \mathbb{N}$ where $n > d$, we define the overdetermined linear system

$$A\boldsymbol{x}^* = \boldsymbol{b} + \boldsymbol{\varepsilon}^*, \qquad A = \begin{pmatrix} \boldsymbol{a}_1^\mathsf{T} \\ \vdots \\ \boldsymbol{a}_n^\mathsf{T} \end{pmatrix} \in \mathbb{R}^{n\times d}, \qquad \boldsymbol{x}^* \in \mathbb{R}^d, \qquad \boldsymbol{b}, \boldsymbol{\varepsilon}^* \in \mathbb{R}^n, \tag{1.1}$$

and we assume that the rows of A are normalized for notational convenience:

$$\forall i \in \{1, ..., n\}, \quad \left\|\boldsymbol{a}_i^\mathsf{T}\right\|_2 = 1. \tag{1.2}$$

(Note that throughout this work, we use the asterisk $(*)$ to denote optimality, not the matrix transpose.)

We also assume that the noise vector $\boldsymbol{\varepsilon}^*$ is sufficiently sparse to ensure that the noiseless rows of the system alone would guarantee a unique solution. That is, fix a "clean" set $\mathcal{C} \subset \{1, .., n\}$ with $|\mathcal{C}| > d$, and assume $\boldsymbol{\varepsilon}_i^* = 0$ for all $i \in \mathcal{C}$. Because $|\mathcal{C}| > d$, we may assume that the matrix A restricted to its clean rows, which we call $A_\mathcal{C}$, has full column rank.

1.2. Kaczmarz Methods

Given A and $\boldsymbol{b}$ (but not $\boldsymbol{\varepsilon}$), we seek to learn $\boldsymbol{x}^*$ using Kaczmarz methods. That is, we begin with an initial guess $\boldsymbol{x}^{(0)} \in \mathbb{R}^d$, and we iteratively refine it by projecting the current solution $\boldsymbol{x}^{(k)}$ onto problem constraints (rows of the system in (1.1)) selected by some criterion. That is, if we select row i_k of the system following the kth iteration, we update $\boldsymbol{x}^{(k)}$ as follows:

$$\begin{aligned} \boldsymbol{x}^{(k+1)} &= \boldsymbol{x}^{(k)} + \frac{b_{i_k} - \left\langle \boldsymbol{a}_{i_k}, \boldsymbol{x}^{(k)} \right\rangle}{\left\|\boldsymbol{a}_{i_k}\right\|^2} \boldsymbol{a}_{i_k} \\ &= \boldsymbol{x}^{(k)} - \varepsilon_{i_k}^{(k)} \boldsymbol{a}_{i_k}, \qquad \text{where we define } \boldsymbol{\varepsilon}^{(k)} = \begin{pmatrix} \varepsilon_1^{(k)} \\ \vdots \\ \varepsilon_n^{(k)} \end{pmatrix} := A\boldsymbol{x}^{(k)} - \boldsymbol{b}. \end{aligned} \tag{1.3}$$

Holds because $-\varepsilon_i^{(k)} \boldsymbol{a}_i = \left(b_i - \boldsymbol{a}_i^\mathsf{T} \boldsymbol{x}^{(k)}\right) \boldsymbol{a}_i$.

Note that $\varepsilon_i^{(k)} = \boldsymbol{a}_i^\mathsf{T} \boldsymbol{x}^{(k)} - b_i$.

We hope that $\boldsymbol{x}^{(k)} \to \boldsymbol{x}^*$. Many methods for choosing i_k at each iteration have been proposed (write about this).

1.3. Deriving an Optimal Selection Criterion

We define $\boldsymbol{e}^{(k)}$ to represent the difference between the kth iterate and the ground truth vector $\boldsymbol{x}^*$, and $e^{(k)}$ to represent its squared 2-norm:

$$\boldsymbol{e}^{(k)} = \boldsymbol{x}^{(k)} - \boldsymbol{x}^*, \qquad e^{(k)} = \left\|\boldsymbol{e}^{(k)}\right\|_2^2.$$

In order to accelerate convergence of $\boldsymbol{x}^{(k)}$ to $\boldsymbol{x}^*$, we seek an i_k at each iteration that will minimize the distance of the next estimate to the true solution, $e^{(k)}$. We do this greedily, since we don't think about the whole sequence, we just think about the current iteration.

Using the abbreviations

$$i := i_k, \quad \varepsilon_i := \varepsilon_i^{(k)}, \quad \text{and} \quad \boldsymbol{x} := \boldsymbol{x}^{(k)}. \tag{1.4}$$

we will expand $\boldsymbol{e}^{(k+1)} = \left\|\boldsymbol{x}^{(k+1)} - \boldsymbol{x}^*\right\|$ using (1.3).

First note that

$$\begin{aligned} \boldsymbol{a}_i^\mathsf{T}(\boldsymbol{x} - \boldsymbol{x}^*) &= \left(\boldsymbol{a}_i^\mathsf{T}\boldsymbol{x} - b_i\right) - \left(\boldsymbol{a}_i^\mathsf{T}\boldsymbol{x}^* - b_i\right) \\ &= \varepsilon_i - \varepsilon_i^*. \end{aligned} \tag{1.5}$$

We expand the expression for the convergence of the $(k+1)$th iterate $\boldsymbol{x}^{(k+1)}$ in terms of the kth iteration (using the abbreviations given in (1.4)):

$$\begin{aligned} e^{(k+1)} = \left\|\boldsymbol{x}^{(k+1)} - \boldsymbol{x}^*\right\|^2 &= \left\|\boldsymbol{x} - \varepsilon_i\boldsymbol{a}_i - \boldsymbol{x}^*\right\|^2 \\ &= \left\|(\boldsymbol{x} - \boldsymbol{x}^*) - \varepsilon_i\boldsymbol{a}_i\right\|^2 \\ &= \left\|\boldsymbol{x} - \boldsymbol{x}^*\right\|^2 + \varepsilon_i^2\left\|\boldsymbol{a}_i\right\|^2 - 2\varepsilon_i\boldsymbol{a}_i^\mathsf{T}(\boldsymbol{x} - \boldsymbol{x}^*) \\ &= e^{(k)} + \varepsilon_i^2 - 2\varepsilon_i(\varepsilon_i - \varepsilon_i^*) \qquad \text{by (1.5)} \\ &= e^{(k)} - \varepsilon_i^2 + 2\varepsilon_i\varepsilon_i^* \\ &= e^{(k)} - \varepsilon_i^2 + 2\varepsilon_i\varepsilon_i^* - (\varepsilon_i^*)^2 + (\varepsilon_i^*)^2 \\ &= e^{(k)} - (\varepsilon_i - \varepsilon_i^*)^2 + (\varepsilon_i^*)^2 \\ &= e^{(k)} - M_i \end{aligned} \tag{1.6}$$

where M_i abbreviates $M_i^{(k)}$, the *optimal selection criterion*, which is defined as

$$M_i^{(k)} := \varepsilon_i^2 - 2\varepsilon_i\varepsilon_i^* = (\varepsilon_i - \varepsilon_i^*)^2 - (\varepsilon_i^*)^2. \tag{1.7}$$

Thus, if we seek to minimize the distance between the next estimate $\boldsymbol{x}$ of $\boldsymbol{x}^*$, the optimal formula for selecting i at some fixed iteration is given by

$$i^* = \underset{i\in\{1,\dots,n\}}{\operatorname{argmax}} M_i, \qquad M_i = (\varepsilon_i - \varepsilon_i^*)^2 - (\varepsilon_i^*)^2. \tag{1.8}$$

Write about how (1.6) motivates maximizing M_i.

1.4. Proposing a Teacher-Optimal Selection Criterion

However, in practice, we do not have access to the true solution $\boldsymbol{x}^*$. Say we have some fixed estimate $\hat{\boldsymbol{x}}$ of the true solution $\boldsymbol{x}^*$, and define its associated residual vector as

$$\hat{\boldsymbol{\varepsilon}} \coloneqq A\hat{\boldsymbol{x}} - \boldsymbol{b}. \tag{1.9}$$

Because we hope that $\hat{\boldsymbol{x}}$ can "teach" us something about $\boldsymbol{x}^*$, we call it the *teacher*. We define the *teacher-optimal selection criterion* much as we defined the optimal selection criterion $M_i^{(k)}$, but using the teacher's residuals $\hat{\boldsymbol{\varepsilon}}$:

$$\widehat{M}_i^{(k)} \coloneqq \varepsilon_i^2 - 2\varepsilon_i\hat{\varepsilon}_i = (\varepsilon_i - \hat{\varepsilon}_i)^2 - \hat{\varepsilon}_i^2, \tag{1.10}$$

which we abbreviate as $\widehat{M}_i$ when the iteration index k is clear or arbitrary.

To apply this criterion, at each iteration we select

$$\hat{\imath} = \underset{i \in \{1,\dots,n\}}{\operatorname{argmax}} \widehat{M}_i. \tag{1.11}$$

In Section 2.1, we demonstrate that the teacher-optimal selection criterion $\widehat{M}$ ((1.10)) converges nicely by showing that the optimal criterion at the teacher-optimal index, $M_{\hat{\imath}}$, is not far off from M_{i^*} when the teacher is reasonably accurate (ensure this is again addressed).

1.5. Assumptions on Teacher

Of course, a teacher model that is wildly inaccurate will not yield near-optimal sample selection in (1.11). As we proceed in demonstrating convergence results for teacher-model online batch selection methods, we let τ represent how near the teacher $\hat{\boldsymbol{x}}$ is to the ground truth solution $\boldsymbol{x}^*$:

$$\tau \coloneqq \left\|\hat{\boldsymbol{\varepsilon}} - \boldsymbol{\varepsilon}^*\right\|_2. \tag{1.12}$$

Thus when $\tau = 0$, the teacher is exactly $\boldsymbol{x}^*$.

2. Full-dataset Methods

In this section, we give convergence bounds and computational results for full-dataset teacher-based sample selection methods. That is, at each training iteration, the criterion considers *every* row of the system as a candidate for selection, as in (1.8) and (1.11). In practice, this is often too expensive, and we instead uniformly sample batches of rows from which to select an optimal candidate. We analyze convergence in expectation of batch-sampled methods in Section 3.

2.1. Teacher-Optimal Method

- Theorems for teacher-optimal method (single sample)
  - Main lemmas
  - Diff eq / iterative / new method? bound proof using said lemmas

We demonstrate the exact relationship between the optimal criterion M_i and the teacher-optimal criterion $\widehat{M}_i$.

Recalling (1.11), we select $\hat{\imath} = \operatorname{argmax}_i \widehat{M}_i$ at each iteration, so (1.6) gives $e^{(k+1)} = e^{(k)} - M_{\hat{\imath}}$. In order to bound $e^{(k+1)}$ from above, we then need a lower bound on $M_{\hat{\imath}}$. We know little about the index $\hat{\imath}$—it could be a clean index ($\hat{\imath} \in \mathcal{C}$), or not.

First, we pin down the exact relationship between $\widehat{M}_i$ and M_i at an arbitrary index $i \in \{1, ..., n\}$, using (1.7) and (1.10).

$$\delta_i := \widehat{M}_i - M_i = (\varepsilon_i^2 - 2\varepsilon_i\hat{\varepsilon}_i) - (\varepsilon_i^2 - 2\varepsilon_i\varepsilon_i^*) = 2\varepsilon_i(\varepsilon_i^* - \hat{\varepsilon}_i). \tag{2.1}$$

Using this relationship, we may place a lower bound on $M_{\hat{\imath}}$. Letting $j \in \{1, ..., n\}$ represent any index, we have $\widehat{M}_{\hat{\imath}} \geq \widehat{M}_j$ by the definition of $\hat{\imath}$, so

$$M_{\hat{\imath}} = \widehat{M}_{\hat{\imath}} - \delta_{\hat{\imath}} \geq \widehat{M}_j - \delta_{\hat{\imath}} = M_j + \delta_j - \delta_{\hat{\imath}}. \tag{2.2}$$

This bound applies for any index j, so we may average it for all j in the clean index set $\mathcal{C}$ to derive a bound on $M_{\hat{\imath}}$ in terms of the clean rows of the system and a bias term.

$$M_{\hat{\imath}} \geq \frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}} M_j + \frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\delta_j - \delta_{\hat{\imath}} \geq \underbrace{\frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\varepsilon_j^2}_{\substack{\text{perfect-teacher}\\ \text{gain}}} - \underbrace{\left|\frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\delta_j\right|}_{\substack{\text{teacher}\\ \text{bias on } \mathcal{C}}} - \underbrace{|\delta_{\hat{\imath}}|}_{\substack{\text{teacher}\\ \text{bias on } \hat{\imath}}}. \tag{2.3}$$

Viewing this bound in the context of (1.6), we have

$$e^{(k+1)} = e^{(k)} - M_{\hat{\imath}} \leq e^{(k)} - \frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\varepsilon_j^2 + \left|\frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\delta_j\right| + |\delta_{\hat{\imath}}|.$$

We will further extend this bound so that it may be written in terms of only $e^{(k)}$ and system constants computable before the start of training. This will allow us to apply it recursively to derive a bound on $e^{(k)}$ in terms of only system constants and $e^{(0)}$, the initial error.

2.1.1. Bounding Gain

We begin by bounding the first bracketed term in (2.3) from below. This term is independent of the teacher, so its bound will depend only on the current iterate $\boldsymbol{x}^{(k)}$ (again abbreviated to $\boldsymbol{x}$) and the clean rows of the system.

$$\begin{aligned}\frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\varepsilon_j^2 &= \frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\left(\varepsilon_j-\varepsilon_j^*\right)^2\\ &= \frac{1}{|\mathcal{C}|}\|A_{\mathcal{C}}(\boldsymbol{x}-\boldsymbol{x}^*)\|^2\\ &\geq \frac{\sigma_{\min}(A_{\mathcal{C}})^2}{|\mathcal{C}|}\|\boldsymbol{x}-\boldsymbol{x}^*\|^2\\ &= \frac{\sigma_{\min}(A_{\mathcal{C}})^2}{|\mathcal{C}|}e^{(k)}\end{aligned} \tag{2.4}$$

2.1.2. Bounding Teacher Bias on Clean Data

To bound the second term in (2.3), we show that (natural language description of techniques).

First, observe that for any $j\in\mathcal{C}$, $\delta_j=-2(\boldsymbol{x}-\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{a}_j\boldsymbol{a}_j^\mathsf{T}(\hat{\boldsymbol{x}}-\boldsymbol{x}^*)$ since $\varepsilon_j^*=0$:

$$\begin{aligned}\delta_j &= 2\varepsilon_j\left(\varepsilon_j^*-\hat{\varepsilon}_j\right)\\ &= -2\left(\varepsilon_j-\varepsilon_j^*\right)\left(\hat{\varepsilon}_j-\varepsilon_j^*\right)\\ &= -2(\boldsymbol{x}-\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{a}_j\boldsymbol{a}_j^\mathsf{T}(\hat{\boldsymbol{x}}-\boldsymbol{x}^*).\end{aligned}$$

This allows us to rewrite the teacher bias term as the inner product of two vectors, which we can then bound with Cauchy–Schwarz.

$$\begin{aligned}\left|\frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\delta_j\right| &= \frac{2}{|\mathcal{C}|}\left|\sum_{j\in\mathcal{C}}(\boldsymbol{x}-\boldsymbol{x}^*)^\mathsf{T}\boldsymbol{a}_j\boldsymbol{a}_j^\mathsf{T}(\hat{\boldsymbol{x}}-\boldsymbol{x}^*)\right|\\ &= \frac{2}{|\mathcal{C}|}\left|(\boldsymbol{x}-\boldsymbol{x}^*)^\mathsf{T}\left(\sum_{j\in\mathcal{C}}\boldsymbol{a}_j\boldsymbol{a}_j^\mathsf{T}\right)(\hat{\boldsymbol{x}}-\boldsymbol{x}^*)\right|\\ &= \frac{2}{|\mathcal{C}|}\left|(A_{\mathcal{C}}(\boldsymbol{x}-\boldsymbol{x}^*))^\mathsf{T}A_{\mathcal{C}}(\hat{\boldsymbol{x}}-\boldsymbol{x}^*)\right|\\ &\leq \frac{2}{|\mathcal{C}|}\|A_{\mathcal{C}}(\boldsymbol{x}-\boldsymbol{x}^*)\|\|A_{\mathcal{C}}(\hat{\boldsymbol{x}}-\boldsymbol{x}^*)\| && \text{by Cauchy– Schwarz}\\ &\leq \frac{2}{\sqrt{|\mathcal{C}|}}\|A_{\mathcal{C}}\|\,\|\boldsymbol{x}-\boldsymbol{x}^*\|\,\|\hat{\boldsymbol{\varepsilon}}-\boldsymbol{\varepsilon}^*\|\\ &\leq \frac{2\tau\sigma_{\max}(A_{\mathcal{C}})}{\sqrt{|\mathcal{C}|}}\sqrt{e^{(k)}} && \text{by (1.12)}\\ &\leq 2\tau\sqrt{e^{(k)}} && \text{because } \sigma_{\max}(A_{\mathcal{C}})\leq\|A_{\mathcal{C}}\|_F=\sqrt{|\mathcal{C}|}.\end{aligned}$$

2.1.3. Bounding Teacher Bias on Selected Row

In bounding the last term of (2.3), we must first note that the teacher-optimal index $\hat{\imath}$ may not lie in $\mathcal{C}$, so we cannot make assumptions about the system noise $\varepsilon_{\hat{\imath}}^*$. We instead bound δ_i for an arbitrary index i, whether clean or not. Because $\delta_i=2\varepsilon_i(\varepsilon_i^*-\hat{\varepsilon}_i)$,

$$\begin{aligned}|\delta_i| &= 2|\varepsilon_i||\varepsilon_i^*-\hat{\varepsilon}_i|\\ &\leq 2|\varepsilon_i-\varepsilon_i^*+\varepsilon_i^*||\varepsilon_i^*-\hat{\varepsilon}_i|\\ &\leq 2\Big(\|\boldsymbol{a}_i\|\|\boldsymbol{x}-\boldsymbol{x}^*\|+\|\boldsymbol{\varepsilon}^*\|_\infty\Big)\|\boldsymbol{\varepsilon}^*-\hat{\boldsymbol{\varepsilon}}\|_2\\ &= 2\Big(\sqrt{e^{(k)}}+\|\boldsymbol{\varepsilon}^*\|_\infty\Big)\tau.\end{aligned} \tag{2.5}$$

Thus $|\delta_{\hat{\imath}}| \le 2\left(\sqrt{e^{(k)}} + \|\varepsilon^*\|_\infty\right)\tau$.

2.1.4. Combining Bounds

Putting these three bounds together into (2.3), we derive the desired lower bound on $M_{\hat{\imath}}$.

$$\begin{aligned} M_{\hat{\imath}} &\ge \frac{\sigma_{\min}(A_{\mathcal{C}})^2}{|\mathcal{C}|} e^{(k)} - 2\tau\sqrt{e^{(k)}} - 2\left(\sqrt{e^{(k)}} + \|\varepsilon^*\|_\infty\right)\tau \\ &= \frac{\sigma_{\min}(A_{\mathcal{C}})^2}{|\mathcal{C}|} e^{(k)} - 4\tau\sqrt{e^{(k)}} - 2\|\varepsilon^*\|_\infty\tau. \end{aligned}$$

The error $e^{(k+1)}$ may thus be bounded above by a quadratic in $\sqrt{e^{(k)}}$:

$$e^{(k+1)} = e^{(k)} - M_{\hat{\imath}} \le \left(1 - \sigma_{\min}\frac{(A_{\mathcal{C}})^2}{|\mathcal{C}|}\right) e^{(k)} + 4\tau\sqrt{e^{(k)}} + 2\|\varepsilon^*\|_\infty\tau.$$

We state this result as Proposition 2.1.

Proposition 2.1. *When we select the training index i_k using the teacher-optimal criterion (1.11), the next iteration's error $e^{(k+1)}$ is related to the previous error by the following inequality:*

$$\begin{aligned} e^{(k+1)} &\le (1-\alpha)e^{(k)} + \beta\sqrt{e^{(k)}} + \gamma, \\ \alpha := \frac{\sigma_{\min}(A_{\mathcal{C}})^2}{|\mathcal{C}|}, &\qquad \beta := 4\tau, \qquad \gamma = 2\|\varepsilon^*\|_\infty\tau. \end{aligned} \tag{2.6}$$

Explain the value of this proposition. Note that when $\tau = 0$, or $\hat{\boldsymbol{x}} = \boldsymbol{x}^*$, the error converges to 0 geometrically in spite of the fact that $\varepsilon^* \neq \mathbf{0}$.

2.1.5. Convergence Results

There are many ways to recursively apply Proposition 2.1 to bound the error at an arbitrary iteration using only constants and the initial state of training. In this section, we give three different methods, each of which gives a different bound, and none of which is strictly better than another (justify this claim).

2.1.5.1. Convergence of Teacher-Optimal Selection Criterion (Via Fixed-Point Shift). First, we determine the fixed point of the bound in Proposition 2.1. Let r be the square root of this error fixed point, so that the fixed point equation becomes quadratic:

$$r^2 = (1-\alpha)r^2 + \beta r + \gamma \qquad \Longrightarrow \qquad r = \frac{\beta + \sqrt{\beta^2 + 4\alpha\gamma}}{2\alpha}.$$

(We use the upper root because $\beta - \sqrt{\beta^2 + 4\alpha\gamma} < \beta - \sqrt{\beta^2} = 0$, but $e^{(k)}$ is nonnegative.)

When $\sqrt{e^{(k)}} = r$, Proposition 2.1 gives $\sqrt{e^{(k+1)}} \le \sqrt{e^{(k)}}$; that is, the bound does not loosen or tighten.

Assume $\sqrt{e^{(k)}} \ge r$, and define $d^{(k)} := \sqrt{e^{(k)}} - r$ to be the distance of the square root error from the fixed point, and rewrite Proposition 2.1 as a quadratic in $d^{(k)}$ using $e^{(k)} = \left(d^{(k)} + r\right)^2$. As we have done with $\boldsymbol{x}^{(k)}$, we abbreviate $d^{(k)}$ to d.

$$\begin{aligned} \left(d^{(k+1)} + r\right)^2 = e^{(k+1)} &\le (1-\alpha)(d+r)^2 + \beta(d+r) + \gamma \\ &= (1-\alpha)\left(d^2 + 2dr + r^2\right) + \beta(d+r) + \gamma \\ &= (1-\alpha)d^2 + (2r(1-\alpha) + \beta)d + \underbrace{(1-\alpha)r^2 + \beta r + \gamma}_{r^2} \end{aligned} \tag{2.7}$$

$$\leq (\rho d + r)^2, \qquad \rho := \max\left(\sqrt{1-\alpha}, 1-\alpha+\frac{\beta}{2r}\right). \tag{2.7}$$

Expanding both sides of (2.7), we may derive a bound on $d^{(k+1)}$ in terms of d, as follows:

$$\begin{aligned} \left(d^{(k+1)} + r\right)^2 &\leq (1-\alpha)d^2 + (2r(1-\alpha)+\beta)d + r^2 \\ \implies d^{(k+1)} &\leq \sqrt{(1-\alpha)d^2 + (2r(1-\alpha)+\beta)d + r^2} - r. \end{aligned} \tag{2.8}$$

We seek a constant ρ such that $d^{(k+1)} \leq \rho d$. By assuming $d > 0$, the above inequality imposes a satisfactory ρ.

$$\begin{aligned} d^{(k+1)} = \sqrt{(1-\alpha)d^2 + (2r(1-\alpha)+\beta)d + r^2} - r &\leq \rho d \\ \iff (1-\alpha)d^2 + (2r(1-\alpha)+\beta)d + r^2 &\leq \rho^2 d^2 + 2\rho r d + r^2 \\ \iff 1-\alpha \leq \rho^2 \quad \text{and} \quad 2r(1-\alpha)+\beta &\leq 2\rho r \\ \iff \rho \geq \sqrt{1-\alpha} \quad \text{and} \quad \rho &\geq 1-\alpha+\frac{\beta}{2r} \\ \impliedby \rho := \max\left(\sqrt{1-\alpha},\ 1-\alpha+\frac{\beta}{2r}\right) & \end{aligned}$$

Recursively applying $d^{(k+1)} \leq \rho d$, we may bound d in terms of $d^{(0)}$:

$$d \leq \rho d^{(k-1)} \leq \cdots \leq \rho^k d^{(0)}. \tag{2.9}$$

Thus $d^{(k)} = \sqrt{e^{(k)}} - r$ converges geometrically toward 0. Rewriting this in terms of $e^{(k)}$, we may add r to both sides of (2.9) and square to derive

$$\left(d^{(k)} + r\right)^2 \leq \left(\rho^k d^{(0)} + r\right)^2,$$

which, by application of the definition of $d^{(k)}$, gives

$$e^{(k)} \leq \left(\rho^k \sqrt{e^{(0)} - r} + r\right)^2.$$

2.1.5.2. Convergence of Teacher-Optimal Selection Criterion (Via Peter–Paul). Proposition 2.1 gives a bound on $e^{(k+1)}$ in terms of $e^{(k)}$. We now use it to derive a bound on $e^{(k)}$ in terms of only $e^{(0)}$ and other information available from the start of training.

Theorem 2.2. *Using the teacher-optimal criterion (1.11), the squared (make this more precise) error converges geometrically toward*

$$e_{\text{fixed}} = 2\left(\frac{\beta^2}{\alpha^2} + \frac{\gamma}{\alpha}\right),$$

with

$$e^{(k)} \leq a^k\left(e^{(0)} - e_{\text{fixed}}\right) + e_{\text{fixed}}.$$

where $a := 1 - \alpha/2$. *reorder things, bound* e_{fixed}

2.1.5.3. Convergence of Teacher-Optimal Selection Criterion (Via Differential Equation). To estimate $\frac{\partial}{\partial k} e^{(k)}$, we can use the forward Euler method with step size 1:

$$\frac{\partial}{\partial k} e^{(k)} \approx \frac{e^{(k+1)} - e^{(k)}}{1}.$$

Explain why it's actually not just $\approx$. It's actually an inequality once k is large enough

By (2.6), we may write

$$\begin{aligned} e^{(k+1)} &= (1-\alpha)e^{(k)} + \beta\sqrt{e^{(k)}} + \gamma \\ \implies e^{(k+1)} - e^{(k)} &\leq f\big(e^{(k)}\big) \end{aligned}$$

where

$$f\big(e^{(k)}\big) := -\alpha e^{(k)} + \beta\sqrt{e^{(k)}} + \gamma,$$

so we may write the bound

$$\frac{d}{dk} e^{(k)} \leq f\big(e^{(k)}\big)$$

Define $dy/dt = f(y) = -\alpha y + \beta\sqrt{y} + \gamma$, and note that when $k = t$,

$$\frac{d}{dk} e^{(k)} \leq \frac{dy}{dt} = f(y),$$

so $e^{(k)} \leq y(t)$ when $k = t$.

We begin by seeking an error floor given by this bound. This will occur when $dy/dt = 0$.

$$\begin{aligned} 0 &= \frac{dy}{dt} \\ \implies 0 &= -\alpha y + \beta\sqrt{y} + \gamma \\ \implies 0 &= -\alpha z^2 + \beta z + \gamma \qquad \text{where } z = \sqrt{y} \\ \implies z &= \frac{-\beta - \sqrt{\beta^2 + 4\alpha\gamma}}{-2\alpha} \\ \implies y &= \left(\frac{-\beta - \sqrt{\beta^2 + 4\alpha\gamma}}{-2\alpha}\right)^2 =: y_{\text{floor}} \end{aligned}$$

Thus, given some error $\bar{e} > y_{\text{floor}}$, if $y(t) \leq \bar{e}$, certainly $e^{(k)} \leq \bar{e}$ as well.

Fix $\bar{e} > y_{\text{floor}}$. We seek the T such that $y(t) = \bar{e}$, thereby giving an upper bound on the lowest k such that $e^{(k)} \leq \bar{e}$.

$$\begin{aligned} \frac{dy}{dt} &= f(y(t)) \\ \implies dt &= \frac{dy}{f(y(t))} \\ \implies T &= \int_{y(0)}^{y(T)} \frac{dy}{f(y(t))} \\ &= \int_{e^{(0)}}^{\bar{e}} \frac{dy}{-\alpha y + \beta\sqrt{y} + \gamma} \\ &= \left(\frac{2}{D}\right)\left(r_+ \log\frac{\sqrt{e^{(0)}} - r_+}{\sqrt{\bar{e}} - r_+} - r_- \log\frac{\sqrt{e^{(0)}} - r_-}{\sqrt{\bar{e}} - r_-}\right), \end{aligned}$$

$$D := \sqrt{\beta^2 + 4\alpha\gamma}, \qquad r_\pm = \frac{\beta \pm D}{2\alpha}$$

Refactor into theorem format.

Explain why the integral holds. https://chatgpt.com/share/e/6ace6d75-7e20-83e8-a20a-af4b368e6f7d

2.2. Reducible Holdout Loss Selection

In *Prioritized Training on Points that are Learnable, Worth Learning, and Not Yet Learnt*, Mindermann et al. propose (in the context of deep learning) another sample selection method, *reducible holdout loss selection (RHO-LOSS)*. We may use this method in a modified form[1] for the Kaczmarz method. Using the same notation as in Section 1, we define RHO-LOSS by the selection criterion

$$\hat{R}_i := \varepsilon_i^2 - \hat{\varepsilon}_i^2, \qquad \hat{\imath} = \underset{i \in \{1,\dots,n\}}{\operatorname{argmax}} \hat{R}_i.$$

We begin with an approach similar to that used for the teacher-optimal method in Section 2.1. Again, (1.6) gives $e^{(k+1)} = e^{(k)} - M_{\hat{\imath}}$, so we seek a lower bound on $M_{\hat{\imath}}$ in order to prove convergence of $e^{(k)}$ to within some neighborhood of 0. We begin by characterizing the relationship between $\hat{R}_i$ and M_i at an arbitrary index $i \in \{1, \dots, n\}$ (compare (2.1)):

$$\delta_i := \hat{R}_i - M_i = (\varepsilon_i^2 - \hat{\varepsilon}_i^2) - (\varepsilon_i^2 - 2\varepsilon_i\varepsilon_i^*) = 2\varepsilon_i\varepsilon_i^* - \hat{\varepsilon}_i^2. \tag{2.10}$$

Using this relationship to begin to bound $M_{\hat{\imath}}$, we find that for any index $j \in \{1, \dots, n\}$ (compare (2.2)),

$$M_{\hat{\imath}} = \hat{R}_{\hat{\imath}} - \delta_{\hat{\imath}} \geq \hat{R}_j - \delta_{\hat{\imath}} = M_j + \delta_j - \delta_{\hat{\imath}}. \tag{2.11}$$

Because (2.11) holds for any j, we may average each side of the inequality over every $j \in \mathcal{C}$ as before (compare (2.3)):

$$M_{\hat{\imath}} \geq \frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}} M_j + \frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}} \delta_j - \delta_{\hat{\imath}} \geq \underbrace{\frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}} \varepsilon_j^2}_{\substack{\text{perfect-teacher} \\ \text{gain}}} - \underbrace{\left|\frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}} \delta_j\right|}_{\text{bias on } \mathcal{C}} - \underbrace{|\delta_{\hat{\imath}}|}_{\text{bias on } \hat{\imath}}. \tag{2.12}$$

Note that this inequality is identical to that in (2.3), except in that δ_i is now defined by (2.10).

2.2.1. Bounding Gain

The lower bound on the first bracketed term is given by (2.4).

2.2.2. Bounding Bias on Clean Data

We now give an upper bound on the second term in (2.12). Complete after rewriting analogue in teacher-opt section.

2.2.3. Bounding Bias on Selected Row

As in Section 2.1.3, we now bound $|\delta_i|$ for arbitrary $i \in \{1, \dots, n\}$, since $\hat{\imath}$ may not lie in $\mathcal{C}$.

$$\begin{aligned} |\delta_i| &= \left|2\varepsilon_i\varepsilon_i^* - \hat{\varepsilon}_i^2\right| \\ &= \left|2\varepsilon_i\varepsilon_i^* + (\varepsilon_i^*)^2 - (\varepsilon_i^* - \hat{\varepsilon}_i)^2\right| \\ &\leq \bigstar \end{aligned}$$

[1] Mindermann et al. define RHO-LOSS using a "holdout" dataset, which is simply a subset of the training data reserved for training the teacher model (here, our teacher model is $\hat{\boldsymbol{x}}$). Because we assume that our teacher is an arbitrary fixed vector and do not discuss the "training" of our teacher, we instead define the holdout loss using the teacher residual.

- Theorems for RHO-LOSS (single sample)
  - Modification of teacher-optimal lemmas for RHO-LOSS if possible?
  - Bound proof

2.3. Empirical Results

- Computational results for single-sample methods, comparison to (probably very loose) bound
  - Gaussian system
  - Real datasets
  - Clock-time comparison to Randomized Kaczmarz? RK should win on large datasets probably? Then batch-sampled opt method should blow RK out of the water

3. Batch-sampled Methods

teststets

- Proof of convergence in expectation for teacher-optimal method
  - Lemmas
  - Proof from lemmas (unless the proof is too complex for that?)
- Proof of convergence in expectation for RHO-LOSS also?
- Computational results for batch-sampled teacher-based Kaczmarz

4. Appendix

- Prove lemmas for each section

Proof of Theorem 2.2. We begin with an application of the Peter-Paul inequality to absorb the $\sqrt{e^{(k)}}$ term of (2.6) into the $e^{(k)}$ term. For any $\eta > 0$,

$$\beta\sqrt{e^{(k)}} \leq \frac{\beta^2}{2\eta} + \frac{\eta e^{(k)}}{2}.$$

To ensure that $1 - \alpha + \eta/2 < 1$, we set $\eta = \alpha$, which gives the following relaxation of (2.6) in terms of only $e^{(k)}$ and constants:

$$e^{(k+1)} \leq \left(1 - \frac{\alpha}{2}\right)e^{(k)} + \frac{\beta^2}{\alpha} + \gamma = ae^{(k)} + c, \qquad a := 1 - \frac{\alpha}{2}, \quad c := \frac{\beta^2}{\alpha} + \gamma. \tag{4.1}$$

Note that the fixed point e_{fixed} of this inequality is

$$e_{\text{fixed}} = 2\left(\frac{\beta^2}{\alpha^2} + \frac{\gamma}{\alpha}\right),$$

since

$$e_{\text{fixed}} = e_{\text{fixed}} - \left(\frac{\beta^2}{\alpha} + \gamma\right) + \frac{\beta^2}{\alpha} + \gamma = ae_{\text{fixed}} + c,$$

and it follows that

$$e_{\text{fixed}} = \frac{c}{1-a}.$$

We now derive a bound on $e^{(k)}$ in terms of only $e^{(0)}$ and constants. By repeated application of (4.1), we have

$$e^{(k)} \leq ae^{(k-1)} + c \leq a\big(ae^{(k-2)} + c\big) + c \leq \cdots \leq a^k e^{(0)} + \sum_{\ell=0}^{k-1} a^\ell c.$$

By the finite geometric series formula, $\sum_{\ell=0}^{k-1} a^\ell c = \frac{c(1-a^k)}{1-a}$, so

$$\begin{aligned} e^{(k)} &\leq a^k e^{(0)} + \frac{c\big(1-a^k\big)}{1-a} \\ &= a^k e^{(0)} + \frac{c}{1-a} - \frac{c}{1-a}a^{(k)} \\ &= a^k\big(e^{(0)} - e_{\text{fixed}}\big) + e_{\text{fixed}}. \end{aligned}$$

□